Question # 2 :-

$$= Q = \begin{bmatrix} 1.5 & 3 \\ 2 & 4 \\ 2.2 & 4.4 \\ 2.4 & 4.4 \end{bmatrix}, \quad K = \begin{bmatrix} 1.1 & 2.2 \\ 1 & 3 \\ 1 & 3 \\ 1.1 & 3 \end{bmatrix}, \quad V = \begin{bmatrix} 0.2 & 0.3 \\ 2 & 0.3 \\ 2.2 & 0.3 \\ 2.2 & 0.3 \end{bmatrix}$$

$$= QK^T = \begin{bmatrix} 1.5 & 3 \\ 2 & 4 \\ 2.2 & 4.4 \\ 2.4 & 4.4 \end{bmatrix} \begin{bmatrix} 1.1 & 1 & 1 & 1.1 \\ 2.2 & 3 & 3 & 3 \end{bmatrix}$$

$$= QK^T = \begin{bmatrix} 1.5\times1.1 + 3\times3 & 1.5\times1 + 3\times3 & 1.5\times1 + 3\times3 & 1.5\times1.1 + 3\times3 \\ 2\times1.1 + 4\times3 & 2\times1 + 4\times3 & 2\times1 + 4\times3 & 2\times1.1 + 4\times3 \\ 2.2\times1.1 + 4.4\times1 & 2.2\times1 + 4.4\times3 & 2.2\times1 + 4.4\times3 & 2.2\times1.1 + 4.4\times3 \\ 2.4\times1.1 + 4.4\times2.2 & 2.4\times1 + 4.4\times3 & 2.4\times1 + 4.4\times3 & 2.4\times1 + 4.4\times3 \end{bmatrix}$$

$$= QK^T = \begin{bmatrix} 8.25 & 10.5 & 10.5 & 10.65 \\ 11 & 14 & 14 & 14.2 \\ 12.1 & 15.4 & 15.4 & 15.62 \\ 12.32 & 15.6 & 15.6 & 15.84 \end{bmatrix}$$

$$= \sqrt{4} = 2$$

$$= \frac{QK^T}{\sqrt{2}} = \begin{bmatrix} 5.83 & 7.42 & 7.42 & 7.53 \\ 7.78 & 9.90 & 9.90 & 10.04 \\ 8.56 & 10.89 & 10.89 & 11.04 \\ 8.71 & 11.03 & 11.03 & 11.20 \end{bmatrix}$$

$$= \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) =$$

$= \text{Row1}\ \Sigma_j e^{x_j} : e^{5.83} + e^{7.42} + e^{7.42} + e^{7.53} = 5542.7$

$= \text{Row2}\ \Sigma_j e^{x_j} : e^{7.78} + e^{9.90} + e^{9.90} + e^{10.04} = 65178.4$

$= \text{Row3}\ \Sigma_j e^{x_j} : e^{8.56} + e^{10.89} + e^{10.89} + e^{11.04} = 174810.9$

$= \text{Row4}\ \Sigma_j e^{x_j} : e^{8.71} + e^{11.03} + e^{11.03} + e^{11.20} = 202588.8$

$$= \begin{bmatrix} e^{5.83}/5542.7 & e^{7.42}/5542.7 & e^{7.42}/5542.7 & e^{7.53}/5542.7 \\ e^{7.78}/65178.4 & e^{9.90}/65178.4 & e^{9.90}/65178.4 & e^{10.04}/65178.4 \\ e^{8.56}/174810.9 & e^{10.89}/174810.9 & e^{10.89}/174810.9 & e^{11.04}/174810.9 \\ e^{8.71}/202588.8 & e^{11.03}/202588.8 & e^{11.03}/202588.8 & e^{11.20}/202588.8 \end{bmatrix}$$

$$= \begin{bmatrix} 0.061 & 0.302 & 0.302 & 0.335 \\ 0.037 & 0.306 & 0.306 & 0.352 \\ 0.030 & 0.306 & 0.306 & 0.358 \\ 0.030 & 0.305 & 0.305 & 0.361 \end{bmatrix} \times \begin{bmatrix} 0.2 & 0.3 \\ 2 & 0.3 \\ 2.2 & 0.3 \\ 2.2 & 0.3 \end{bmatrix}$$

$$= \begin{bmatrix} 2.02 & 0.3 \\ 2.07 & 0.3 \\ 2.08 & 0.3 \\ 2.08 & 0.3 \end{bmatrix} \begin{matrix} \rightarrow \text{Agentic} \\ \rightarrow \text{AI} \\ \rightarrow \text{emerging} \\ \rightarrow \text{course} \end{matrix}$$

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